

Rajat Saxena

✉ Rajat.Saxena@campus.lmu.de | 🌐 Portfolio

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ABOUT

Master's student at LMU Munich with experience developing simulations in theoretical physics and astrophysics. My research interests lie in statistical mechanics and the physics of machine learning. I am also passionate about science outreach, philosophy and history.

EDUCATION

- **MSc in Physics** October 2024 - Present
Ludwig Maximilian University (LMU) of Munich; **Grade: 2.63** (1 Highest, 5 Fail)
Advisor: Dr. Steffen Rulands, Professor, LMU Munich, Germany
- **BSc Blended Physics** June 2024
Savitribai Phule Pune University, **CGPA: 9.32** (10 Highest, 4 Fail)
Advisor: Dr Deepak Dhar, Distinguished Professor Emeritus, IISER - Pune Pune, India

RESEARCH EXPERIENCE

- **Universitäts-Sternwarte München** March 2025 - Present
Research Intern
Mentor: Dr Tilman Birnstiel, Professor
 - Developed N-body simulations to investigate the gravitational collapse of a dust cloud in the presence of a central star and a pre-formed gas disk.
 - Modelled particle trajectories coupled with gas dynamics during protostar formation, with integration of hydrodynamic snapshots (from [Tārā](#)) to interpolate density, velocity, and temperature fields beyond simplified disk models.
 - Summarised the group's research for Astrobites, translating complex results into accessible articles. Available: [\[🌐\]](#)
- **Indian Institute of Science Education and Research (IISER) – Pune** August 2023 - May 2024
Bachelor Thesis
Mentor: Dr Deepak Dhar, Distinguished Professor Emeritus
 - Studied foundational statistical mechanics and phase transitions through standard texts to build a strong theoretical base.
 - Developed a Python simulation of a 1D zero-temperature random field Ising model to investigate discrete magnetisation jumps in ferromagnetic materials with impurities and defects.
 - Analysed the impact of lattice size, random field standard deviation, and probability distributions on magnetisation curves, quantifying parameter-dependent behaviour.
- **Inter-University Centre for Astronomy and Astrophysics (IUCAA)** June 2023 - July 2023
Summer Intern
Mentor: Mr Ashish Mhaske, Scientific and Technical Officer
 - Led a student team to measure 21-cm neutral hydrogen power spectrum across different Galactic longitudes using a conical horn antenna.
 - Based on the Doppler-shifted HI line profiles, obtained the Milky Way Rotation Curve.
 - Designed and implemented a Python data-analysis pipeline to process source and calibration data.
- **National Centre for Radio Astrophysics (NCRA) - TIFR** September 2022 – April 2023
Research Intern
Mentor: Dr Yashwant Gupta, Distinguished Professor and Centre Director
 - Developed Python code for folding and binning time-series data to generate pulsar profiles for the Crab Pulsar using uGMRT data.
 - Implemented C routines to define on/off gates, enabling precise separation of pulsar on-pulse and off-pulse emission intervals.
 - Applied pulsar timing, dedispersion techniques, and correlator analysis, leveraging software tools such as TEMPO2 and GMRT Pulsar Tool (gptool).

OUTREACH EXPERIENCE

• **LIGO – India Education and Public Outreach (LI-EPO), IUCAA**

January 2023 – July 2024

Intern

Mentor: Dr Debarati Chatterjee, Associate Professor and Chair LI-EPO

- Developed outreach content and social media posts to communicate LIGO India's mission, maintaining interferometer demonstrations for public engagement.
- Conducted awareness campaigns across Indian colleges, promoting the upcoming observatory.
- Designed and led an introductory Python workshop for undergraduates, and assisted in organising Astronomy Teacher Training Workshops and local outreach at the Hingoli LIGO India site.

• **Science Popularisation Centre (SciPop), IUCAA**

Volunteer

- Operated telescopes and led stargazing sessions for school students and the general public, training new volunteers in telescope operation.

MENTORSHIP EXPERIENCE

- **Teaching Assistant at Quest Classes, Pune** (Aug–Oct 2024): Taught 12th-grade Physics (optics) and 11th-grade Mathematics (inequalities, combinatorics, binomial theorem).

- **Co-founder & Research Head, IDSS Space Society** (Oct 2021 – Jul 2023): Organised guest lectures, stargazing nights, school outreach programs, and space-themed MUNs; produced educational social media content. Recognised as a "Registered Space Tutor of ISRO" by CBPO, ISRO.

SELECTED CONFERENCES

Oral Presentations

- **30th Young Scientists' Conference on Astronomy and Space Physics**, National University of Kyiv *April 2024*
- **10th National Student Symposium on Physics**, Indian Association Physics Teachers (IAPT) *October 2023*

Poster Presentations

- **Symposium on Magnetism and Spintronics**, Indian Institute of Technology - Bombay *July 2024*
- **National Space Science Symposium**, Indian Space Research Organization (ISRO) *February 2024*

RELEVANT WORKSHOPS

- **German Italian Physics Exchange**, German Physical Society & Associazione Italiana Studenti di Fisica *Sept 2025*
- **Summer University for Plasma Physics and Fusion Research**, Max Planck Institute for Plasma Physics *Sept 2025*
- **Summer School in Theoretical (Astro)Physics**, St Xavier's College & IUCAA *June 2024*
- **Course on Nuclear Experiments**, HBCSE-TIFR *March 2024*
- **Workshop on Data Science in Astronomy**, IUCAA *December 2023*

SKILLS

- **Programming Languages:** Python, Mathematica, Latex
- **Libraries:** Numpy, Scipy, Astropy, REBOUND, Matplotlib, Pandas
- **ML Libraries:** Keras/TensorFlow, Sklearn
- **Telescope Handling:** Dobsonian & Newtonian telescopes (EQ/Altazimuth mounts)
- **Softwares:** LAMMPS, ORCA, Astrometrica
- **Languages:** English (IELTS: 7.5), Hindi (Native), Marathi (Professional)

AWARDS AND ACHIEVEMENTS

• **Exceptional Accomplishments in Physics and Science Communication**

March 2024

Interdisciplinary School of Science, Savitribai Phule Pune University

- Conferred by the Hon. Vice Chancellor in recognition of outstanding academic performance and science communication initiatives.
- Acknowledged as a significant departmental honor highlighting excellence in both research and outreach.

• **Telescope Building Grant (INR 60,000)**

2022–23

Savitribai Phule Pune University

- Received university funding to design and construct a Newtonian telescope.

PROFESSIONAL MEMBERSHIPS

- **Student Member of "German Physical Society"** 2025 - Present
- **Student Member of "Indian Physics Association"** 2024 - Present

PUBLICATIONS

- [1] Saxena, Rajat. (2024). Introduction to Statistical Mechanics and Simulation of Random Field Ising Model. 10.13140/RG.2.2.34379.81443. Available: [\[🌐\]](#)
- [2] **Saxena, Rajat** & Padalkar, Gauri & Patil, Yash & Sherkar, Shreenath & Satam, Harsh & Shah, Manasvi & Rajopadhye, Chinmayee. (2024). Galaxy Rotation Curve Measurements using a Horn Antenna. Available: [\[🌐\]](#)